# Evaluation runs: 2pc-lb (two-pc-lbinria)

generated 2026-11-08 from brain-data e1e40f443 2026-11-08 13:42:54 +0100

Every campaign run on this benchmark, one row per instance (variant, specification, communication model, size) and engine; the newest run of an instance wins. Expected verdicts (the rule file): no entry in the rule file (the README states the expectation). Blue rows are at the current pins (TraceForge 367cdc6, whose predecessor 9a0629e is equivalent for every engine but the join; merge join 367cdc6; JMC 3c43b2d2): a rerun gives the same result. Red rows are superseded (an older pin, or the in-memory join that the merge join replaces) and their rerun is pending. A planned instance without a measurement is not a row: it is listed under "Not run". $G_{ref}$, $G_{cand}$ = the graphs of the reference and of the implementation (the stateless engine's own count of specification graphs explored, repeats included, is $G_{ref}^{expl}$); $G_x$ = the implementation graphs no specification graph covers (violating pairs); T = wall seconds; M = peak resident MiB; tier = the run's time and memory limits; * marks a verdict other than the expected one; 🕓 = the time tier was reached; oom = the memory limit was reached.

## stateless

### stateless: pair **correct**, model bag, expected conforms

| size | $G_{spec}$ | $G_{impl}$ | verdict | T | M | tier | sweeps | $G_{ref}^{expl}$ | run |
|---|---|---|---|---|---|---|---|---|---|
| 2 | 15 | 76 | conforms | 0.0 | 2 | 1 h, 250 GB | 7 | 36 | brain03, 2026-11-02 |
| 3 | 54 | 1690 | conforms | 0.1 | 2 | 1 h, 250 GB | 15 | 284 | brain03, 2026-11-02 |
| 4 | 233 | 72346 | conforms | 1.6 | 2 | 1 h, 250 GB | 62 | 8003 | brain03, 2026-11-02 |
| 5 | 1198 | 5234244 | conforms | 43.7 | 28 | 48 h, 250 GB | 65 | 24615 | brain03, 2026-11-07 |
| 6 | 7253 | 576552418 | conforms | 6843.0 | 614 | 48 h, 250 GB | 386 | 1611773 | brain05, 2026-11-07 |

### stateless: pair **counting-bug**, model bag, expected violation

| size | $G_{spec}$ | $G_{impl}$ | $G_x$ | verdict | T | M | tier | sweeps | $G_{ref}^{expl}$ | run |
|---|---|---|---|---|---|---|---|---|---|---|
| 2 | 15 | 60 | 12 | violation | 0.0 | 2 | 1 h, 250 GB | 19 | 216 | brain03, 2026-11-02 |
| 3 | 54 | 826 | 324 | violation | 0.7 | 4 | 1 h, 250 GB | 339 | 17694 | brain03, 2026-11-02 |
| 4 | 233 | 16618 | 9072 | violation | 44.4 | 6 | 1 h, 250 GB | 9136 | 2121572 | brain03, 2026-11-02 |
| 5 | 1198 | 447444 | 291600 | violation | 3150.1 | 12 | 1 h, 250 GB | 291763 | 349481933 | brain03, 2026-11-02 |
| 6 | 7253 | | | 🕓 (1 h) | | | 1 h, 250 GB | | | brain03, 2026-11-02 |

**Not run (stateless).** planned but never scheduled (grid points beyond the ladder): 5 bag correct, 6 bag correct, 6 correct bag, 6 bag counting-bug, 6 bag counting-bug.t16.p.r3, 7 bag correct, 7 bag counting-bug, 7 bag correct.t16.p.r3, 7 bag counting-bug.t16.p.r3, 8 bag counting-bug.

left out by the table rules (ok): 5 correct bag.

left out by the table rules (timeout): 6 correct bag.

skipped: a smaller size had reached its limit: 7 counting-bug bag, 7 correct bag, 8 correct bag, 8 counting-bug bag, 9 counting-bug bag, 9 correct bag, 10 counting-bug bag, 10 correct bag.

## stateful

### stateful: pair **correct**, model bag, expected conforms

| size | $G_{spec}$ | $G_{impl}$ | verdict | T | M | tier | run |
|---|---|---|---|---|---|---|---|
| 2 | 15 | 76 | conforms | 0.0 | 2 | 1 h, 250 GB | brain02, 2026-11-02 |
| 3 | 54 | 1690 | conforms | 0.2 | 2 | 1 h, 250 GB | brain02, 2026-11-02 |
| 4 | 233 | 72346 | conforms | 1.8 | 2 | 1 h, 250 GB | brain02, 2026-11-02 |
| 5 | 1198 | 5234244 | conforms | 46.7 | 29 | 48 h, 250 GB | brain03, 2026-11-07 |
| 6 | 7253 | 576552418 | conforms | 5140.0 | 652 | 48 h, 250 GB | brain04, 2026-11-05 |

### stateful: pair **counting-bug**, model bag, expected violation

| size | $G_{spec}$ | $G_{impl}$ | $G_x$ | verdict | T | M | tier | run |
|---|---|---|---|---|---|---|---|---|
| 2 | 15 | 60 | 12 | violation | 0.0 | 2 | 1 h, 250 GB | brain02, 2026-11-02 |
| 3 | 54 | 826 | 324 | violation | 0.1 | 4 | 1 h, 250 GB | brain02, 2026-11-02 |
| 4 | 233 | 16618 | 9072 | violation | 0.8 | 2 | 1 h, 250 GB | brain02, 2026-11-02 |
| 5 | 1198 | 447444 | 291600 | violation | 6.7 | 2 | 1 h, 250 GB | brain02, 2026-11-02 |
| 6 | 7253 | 15089218 | 10886400 | violation | 142.7 | 40 | 48 h, 250 GB | brain04, 2026-11-07 |
| 7 | 50900 | 610272436 | 468659520 | violation | 7021.0 | 674 | 48 h, 250 GB | brain05, 2026-11-02 |

**Not run (stateful).** planned but never scheduled (grid points beyond the ladder): 6 correct bag, 7 bag correct.t16.p.r3, 7 bag counting-bug.t16.p.r3, 8 bag correct, 8 correct bag, 9 counting-bug bag, 9 correct bag, 10 counting-bug bag, 10 correct bag, 10 bag counting-bug, 11 correct bag, 11 counting-bug bag, 12 correct bag, 12 counting-bug bag, 13 counting-bug bag, 13 correct bag, 14 counting-bug bag.

left out by the table rules (ok): 5 correct bag, 5 bag correct, 6 counting-bug bag.

left out by the table rules (timeout): 6 correct bag, 7 counting-bug bag, 8 bag counting-bug.

skipped: a smaller size had reached its limit: 7 correct bag, 8 correct bag, 8 counting-bug bag, 9 counting-bug bag, 9 correct bag, 10 counting-bug bag, 10 correct bag.

the tool ended with an error (exit-101): 7 bag correct.

left out by the table rules (invalid: foreign load): 9 bag correct, 9 bag counting-bug.

## offline

### offline: pair **correct**, model bag, expected conforms

| size | $G_{spec}$ | $G_{impl}$ | verdict | T | M | tier | export spec | export impl | join | join kind | run |
|---|---|---|---|---|---|---|---|---|---|---|---|
| 2 | 15 | 76 | conforms | 0.2 | 8 | 10 min, 16 GB | 0.0 | 0.2 | 0.0 | in-memory | brain05, 2026-11-02 |
| 3 | 54 | 1690 | conforms | 0.1 | 8 | 10 min, 16 GB | 0.0 | 0.1 | 0.0 | in-memory | brain05, 2026-11-02 |
| 4 | 233 | 72346 | conforms | 1.7 | 8 | 1 h, 250 GB | 0.1 | 1.7 | 0.0 | merge | brain05, 2026-11-03 |
| 5 | 1198 | 5234244 | conforms | 52.0 | 28 | 48 h, 250 GB | 0.1 | 52.0 | 0.0 | merge | brain03, 2026-11-07 |
| 5 | 1198 | 5234244 | conforms | 53.5 | 30 | 1 h, 250 GB | 0.1 | 53.5 | 0.0 | merge | brain05, 2026-11-03 |
| 6 | 7253 | 576552418 | conforms | 5706.0 | 632 | 48 h, 250 GB | 0.3 | 5706.0 | 0.0 | merge | brain04, 2026-11-06 |

### offline: pair **counting-bug**, model bag, expected violation

| size | $G_{spec}$ | $G_{impl}$ | $G_x$ | verdict | T | M | tier | export spec | export impl | join | join kind | run |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 2 | 15 | 60 | 4 | violation | 0.0 | 8 | 10 min, 16 GB | 0.0 | 0.0 | 0.0 | in-memory | brain05, 2026-11-02 |
| 3 | 54 | 826 | 6 | violation | 0.3 | 8 | 10 min, 16 GB | 0.0 | 0.3 | 0.0 | in-memory | brain05, 2026-11-02 |
| 4 | 233 | 16618 | 32 | violation | 0.9 | 8 | 10 min, 16 GB | 0.1 | 0.9 | 0.0 | in-memory | brain05, 2026-11-02 |
| 5 | 1198 | 447444 | 50 | violation | 8.9 | 8 | 1 h, 250 GB | 0.1 | 8.8 | 0.0 | merge | brain05, 2026-11-03 |
| 6 | 7253 | 15089218 | 232 | violation | 142.1 | 38 | 48 h, 250 GB | 0.3 | 142.0 | 0.0 | merge | brain04, 2026-11-07 |
| 6 | 7253 | 15089218 | 232 | violation | 162.4 | 39 | 1 h, 250 GB | 0.5 | 162.4 | 0.0 | merge | brain05, 2026-11-03 |

**Not run (offline).** planned but never scheduled (grid points beyond the ladder): 5 bag correct, 7 bag correct.p.r3, 7 bag counting-bug.p.r3, 8 bag counting-bug.

skipped: a smaller size had reached its limit: 6 correct bag, 7 bag correct, 7 bag counting-bug, 7 counting-bug bag, 7 correct bag, 8 correct bag, 8 counting-bug bag, 9 counting-bug bag, 9 correct bag, 10 counting-bug bag, 10 correct bag.

## cross offline, Rust implementation against the Java specification

### cross offline, Rust implementation against the Java specification: pair **correct**, model bag, expected conforms

| size | $G_{spec}$ | $G_{impl}$ | verdict | T | M | tier | export spec | export impl | join | join kind | run |
|---|---|---|---|---|---|---|---|---|---|---|---|
| 2 | 15 | 76 | conforms | 3.1 | 162 | 10 min, 16 GB | 3.1 | 0.2 | 0.0 | in-memory | brain01, 2026-11-02 |
| 3 | 54 | 1690 | conforms | 4.1 | 172 | 10 min, 250 GB | 4.0 | 0.1 | 0.1 | merge | brain01, 2026-11-03 |
| 4 | 233 | 72346 | conforms | 4.4 | 185 | 10 min, 250 GB | 4.4 | 1.7 | 0.0 | merge | brain01, 2026-11-03 |

### cross offline, Rust implementation against the Java specification: pair **counting-bug**, model bag, expected violation

| size | $G_{spec}$ | $G_{impl}$ | $G_x$ | verdict | T | M | tier | export spec | export impl | join | join kind | run |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 2 | 15 | 60 | 4 | violation | 3.1 | 162 | 10 min, 16 GB | 3.1 | 0.0 | 0.0 | in-memory | brain01, 2026-11-02 |
| 3 | 54 | 826 | 6 | violation | 4.2 | 172 | 10 min, 250 GB | 4.0 | 0.3 | 0.2 | merge | brain01, 2026-11-03 |
| 4 | 233 | 16618 | 32 | violation | 4.5 | 185 | 10 min, 250 GB | 4.4 | 0.9 | 0.1 | merge | brain01, 2026-11-03 |

**Not run (cross offline, Rust implementation against the Java specification).** planned but never scheduled (grid points beyond the ladder): 5 bag correct, 5 bag counting-bug, 6 bag correct, 6 bag counting-bug, 7 bag correct, 7 bag counting-bug, 8 bag counting-bug.

## cross online, Rust implementation (TraceForge) against the Java specification export

### cross online, Rust implementation (TraceForge) against the Java specification export: pair **correct**, model bag, expected conforms

| size | $G_{spec}$ | $G_{impl}$ | verdict | T | M | tier | first miss ms | run |
|---|---|---|---|---|---|---|---|---|
| 2 | 15 | 76 | conforms | 3.1 | 158 | 1 h, 250 GB | | brain01, 2026-11-02 |
| 3 | 54 | 1690 | conforms | 3.2 | 161 | 1 h, 250 GB | | brain01, 2026-11-02 |
| 4 | 233 | 72346 | conforms | 6.0 | 170 | 1 h, 250 GB | | brain01, 2026-11-02 |

### cross online, Rust implementation (TraceForge) against the Java specification export: pair **counting-bug**, model bag, expected violation

| size | $G_{spec}$ | $G_{impl}$ | $G_x$ | verdict | T | M | tier | first miss ms | run |
|---|---|---|---|---|---|---|---|---|---|
| 2 | 15 | 60 | 12 | violation | 3.1 | 158 | 1 h, 250 GB | 2 | brain01, 2026-11-02 |
| 3 | 54 | 826 | 324 | violation | 3.2 | 161 | 1 h, 250 GB | 2 | brain01, 2026-11-02 |
| 4 | 233 | 16618 | 9072 | violation | 4.6 | 170 | 1 h, 250 GB | 14 | brain01, 2026-11-02 |

**Not run (cross online, Rust implementation (TraceForge) against the Java specification export).** planned but never scheduled (grid points beyond the ladder): 5 bag correct, 6 bag correct, 6 bag counting-bug, 7 bag correct, 7 bag counting-bug, 8 bag counting-bug.

## twins offline, Rust implementation against its Java twin

### twins offline, Rust implementation against its Java twin: pair **correct**, model bag, expected conforms

| size | $G_{spec}$ | $G_{impl}$ | verdict | T | M | tier | export spec | export impl | join | join kind | run |
|---|---|---|---|---|---|---|---|---|---|---|---|
| 2 | 76 | 76 | conforms | 30.3 | 1892 | 4 h, 250 GB | 30.3 | 0.2 | 0.0 | merge | brain01, 2026-11-03 |
| 3 | 1690 | 1690 | conforms | 9.1 | 541 | 10 min, 250 GB | 9.1 | 0.1 | 0.1 | merge | brain01, 2026-11-03 |
| 4 | 72346 | 72346 | conforms | 225.9 | 3049 | 10 min, 250 GB | 225.8 | 1.7 | 0.1 | merge | brain01, 2026-11-03 |

### twins offline, Rust implementation against its Java twin: pair **counting-bug**, model bag, expected conforms

| size | $G_{spec}$ | $G_{impl}$ | verdict | T | M | tier | export spec | export impl | join | join kind | run |
|---|---|---|---|---|---|---|---|---|---|---|---|
| 2 | 60 | 60 | conforms | 12.1 | 1684 | 4 h, 250 GB | 12.0 | 0.0 | 0.0 | merge | brain01, 2026-11-03 |
| 3 | 826 | 826 | conforms | 6.8 | 345 | 10 min, 250 GB | 6.7 | 0.3 | 0.1 | merge | brain01, 2026-11-03 |
| 4 | 16618 | 16618 | conforms | 92.2 | 2420 | 10 min, 250 GB | 92.2 | 0.9 | 0.1 | merge | brain01, 2026-11-03 |

### twins offline, Rust implementation against its Java twin: pair **counting-bug**, model bag, expected violation

| size | $G_{spec}$ | $G_{impl}$ | $G_x$ | verdict | T | M | tier | export spec | export impl | join | join kind | run |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 2 | 76 | 60 | 4 | violation | 30.3 | 1892 | 4 h, 250 GB | 30.3 | 0.0 | 0.0 | merge | brain01, 2026-11-03 |
| 3 | 1690 | 826 | 6 | violation | 9.1 | 541 | 10 min, 250 GB | 9.1 | 0.3 | 0.0 | merge | brain01, 2026-11-03 |
| 4 | 72346 | 16618 | 32 | violation | 225.9 | 3049 | 10 min, 250 GB | 225.8 | 0.9 | 0.1 | merge | brain01, 2026-11-03 |

**Not run (twins offline, Rust implementation against its Java twin).** planned but never scheduled (grid points beyond the ladder): 5 bag correct, 5 bag counting-bug, 6 bag correct, 6 bag counting-bug, 7 bag counting-bug.

## twins offline, Java implementation against its Rust twin

### twins offline, Java implementation against its Rust twin: pair **correct**, model bag, expected conforms

| size | $G_{spec}$ | $G_{impl}$ | verdict | T | M | tier | export spec | export impl | join | join kind | run |
|---|---|---|---|---|---|---|---|---|---|---|---|
| 2 | 76 | 76 | conforms | 30.3 | 1892 | 4 h, 250 GB | 0.2 | 30.3 | 0.0 | merge | brain01, 2026-11-03 |
| 3 | 1690 | 1690 | conforms | 9.4 | 541 | 10 min, 250 GB | 0.1 | 9.1 | 0.3 | merge | brain01, 2026-11-03 |
| 4 | 72346 | 72346 | conforms | 225.8 | 3049 | 10 min, 250 GB | 1.7 | 225.8 | 0.1 | merge | brain01, 2026-11-03 |

### twins offline, Java implementation against its Rust twin: pair **counting-bug**, model bag, expected conforms

| size | $G_{spec}$ | $G_{impl}$ | verdict | T | M | tier | export spec | export impl | join | join kind | run |
|---|---|---|---|---|---|---|---|---|---|---|---|
| 2 | 60 | 60 | conforms | 12.0 | 1684 | 4 h, 250 GB | 0.0 | 12.0 | 0.0 | merge | brain01, 2026-11-03 |
| 3 | 826 | 826 | conforms | 6.7 | 345 | 10 min, 250 GB | 0.3 | 6.7 | 0.1 | merge | brain01, 2026-11-03 |
| 4 | 16618 | 16618 | conforms | 92.3 | 2420 | 10 min, 250 GB | 0.9 | 92.2 | 0.1 | merge | brain01, 2026-11-03 |

### twins offline, Java implementation against its Rust twin: pair **counting-bug**, model bag, expected violation

| size | $G_{spec}$ | $G_{impl}$ | $G_x$ | verdict | T | M | tier | export spec | export impl | join | join kind | run |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 2 | 76 | 60 | 4 | violation | 12.0 | 1684 | 4 h, 250 GB | 0.2 | 12.0 | 0.0 | merge | brain01, 2026-11-03 |
| 3 | 1690 | 826 | 6 | violation | 6.8 | 345 | 10 min, 250 GB | 0.1 | 6.7 | 0.1 | merge | brain01, 2026-11-03 |
| 4 | 72346 | 16618 | 32 | violation | 92.4 | 2420 | 10 min, 250 GB | 1.7 | 92.2 | 0.2 | merge | brain01, 2026-11-03 |

**Not run (twins offline, Java implementation against its Rust twin).** planned but never scheduled (grid points beyond the ladder): 5 bag correct, 5 bag counting-bug, 6 bag correct, 6 bag counting-bug, 7 bag counting-bug.

## context-free twins offline, Rust implementation against the context-free Java port

### context-free twins offline, Rust implementation against the context-free Java port: pair **correct**, model bag, expected conforms

| size | $G_{spec}$ | $G_{impl}$ | verdict | T | M | tier | export spec | export impl | join | join kind | run |
|---|---|---|---|---|---|---|---|---|---|---|---|
| 2 | 76 | 76 | conforms | 4.8 | 164 | 4 h, 250 GB | 4.8 | 0.2 | 0.0 | merge | brain01, 2026-11-03 |
| 3 | 1690 | 1690 | conforms | 11.6 | 2189 | 10 min, 250 GB | 11.5 | 0.1 | 0.1 | merge | brain01, 2026-11-03 |
| 4 | 72346 | 72346 | conforms | 276.6 | 3147 | 10 min, 250 GB | 276.5 | 1.7 | 0.0 | merge | brain01, 2026-11-03 |

### context-free twins offline, Rust implementation against the context-free Java port: pair **counting-bug**, model bag, expected conforms

| size | $G_{spec}$ | $G_{impl}$ | verdict | T | M | tier | export spec | export impl | join | join kind | run |
|---|---|---|---|---|---|---|---|---|---|---|---|
| 2 | 60 | 60 | conforms | 5.1 | 168 | 4 h, 250 GB | 5.1 | 0.0 | 0.0 | merge | brain01, 2026-11-03 |
| 3 | 826 | 826 | conforms | 8.1 | 325 | 10 min, 250 GB | 7.9 | 0.3 | 0.2 | merge | brain01, 2026-11-03 |
| 4 | 16618 | 16618 | conforms | 91.7 | 2560 | 10 min, 250 GB | 91.7 | 0.9 | 0.0 | merge | brain01, 2026-11-03 |

### context-free twins offline, Rust implementation against the context-free Java port: pair **counting-bug**, model bag, expected violation

| size | $G_{spec}$ | $G_{impl}$ | $G_x$ | verdict | T | M | tier | export spec | export impl | join | join kind | run |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 2 | 76 | 60 | 4 | violation | 4.8 | 164 | 4 h, 250 GB | 4.8 | 0.0 | 0.0 | merge | brain01, 2026-11-03 |
| 3 | 1690 | 826 | 6 | violation | 12.1 | 2190 | 10 min, 250 GB | 11.5 | 0.3 | 0.6 | merge | brain01, 2026-11-03 |
| 4 | 72346 | 16618 | 32 | violation | 276.6 | 3147 | 10 min, 250 GB | 276.5 | 0.9 | 0.0 | merge | brain01, 2026-11-03 |

**Not run (context-free twins offline, Rust implementation against the context-free Java port).** planned but never scheduled (grid points beyond the ladder): 5 bag correct, 5 bag counting-bug, 6 bag correct, 6 bag counting-bug, 7 bag counting-bug.

## context-free twins offline, Java port against the Rust implementation

### context-free twins offline, Java port against the Rust implementation: pair **correct**, model bag, expected conforms

| size | $G_{spec}$ | $G_{impl}$ | verdict | T | M | tier | export spec | export impl | join | join kind | run |
|---|---|---|---|---|---|---|---|---|---|---|---|
| 2 | 76 | 76 | conforms | 4.8 | 164 | 4 h, 250 GB | 0.2 | 4.8 | 0.0 | merge | brain01, 2026-11-05 |
| 3 | 1690 | 1690 | conforms | 11.8 | 2189 | 10 min, 250 GB | 0.1 | 11.5 | 0.3 | merge | brain01, 2026-11-03 |
| 4 | 72346 | 72346 | conforms | 276.9 | 3147 | 10 min, 250 GB | 1.7 | 276.5 | 0.3 | merge | brain01, 2026-11-03 |

### context-free twins offline, Java port against the Rust implementation: pair **counting-bug**, model bag, expected conforms

| size | $G_{spec}$ | $G_{impl}$ | verdict | T | M | tier | export spec | export impl | join | join kind | run |
|---|---|---|---|---|---|---|---|---|---|---|---|
| 2 | 60 | 60 | conforms | 5.1 | 168 | 4 h, 250 GB | 0.0 | 5.1 | 0.0 | merge | brain01, 2026-11-05 |
| 3 | 826 | 826 | conforms | 7.9 | 325 | 10 min, 250 GB | 0.3 | 7.9 | 0.0 | merge | brain01, 2026-11-03 |
| 4 | 16618 | 16618 | conforms | 92.1 | 2560 | 10 min, 250 GB | 0.9 | 91.7 | 0.4 | merge | brain01, 2026-11-03 |

### context-free twins offline, Java port against the Rust implementation: pair **counting-bug**, model bag, expected violation

| size | $G_{spec}$ | $G_{impl}$ | $G_x$ | verdict | T | M | tier | export spec | export impl | join | join kind | run |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 2 | 76 | 60 | 4 | violation | 5.1 | 168 | 4 h, 250 GB | 0.2 | 5.1 | 0.0 | merge | brain01, 2026-11-05 |
| 3 | 1690 | 826 | 6 | violation | 8.3 | 325 | 10 min, 250 GB | 0.1 | 7.9 | 0.4 | merge | brain01, 2026-11-03 |
| 4 | 72346 | 16618 | 32 | violation | 91.7 | 2560 | 10 min, 250 GB | 1.7 | 91.7 | 0.0 | merge | brain01, 2026-11-03 |

**Not run (context-free twins offline, Java port against the Rust implementation).** planned but never scheduled (grid points beyond the ladder): 5 bag counting-bug, 6 bag counting-bug, 6 bag correct, 7 bag counting-bug.
refused (an input missing at run time): 5 bag correct.

## context-free twins online, Java port (JMC) against the Rust implementation's export

### context-free twins online, Java port (JMC) against the Rust implementation's export: pair **correct**, model bag, expected conforms

| size | $G_{spec}$ | $G_{impl}$ | verdict | T | M | tier | first miss ms | run |
|---|---|---|---|---|---|---|---|---|
| 2 | 76 | 76 | conforms | 3.5 | 169 | 1 h, 250 GB | | brain01, 2026-11-02 |
| 3 | 1690 | 1690 | conforms | 15.3 | 1861 | 1 h, 250 GB | | brain01, 2026-11-03 |
| 4 | 72346 | 72346 | conforms | 324.2 | 3141 | 1 h, 250 GB | | brain01, 2026-11-03 |

### context-free twins online, Java port (JMC) against the Rust implementation's export: pair **counting-bug**, model bag, expected conforms

| size | $G_{spec}$ | $G_{impl}$ | verdict | T | M | tier | first miss ms | run |
|---|---|---|---|---|---|---|---|---|
| 2 | 60 | 60 | conforms | 3.4 | 167 | 1 h, 250 GB | | brain01, 2026-11-02 |
| 3 | 826 | 826 | conforms | 12.0 | 1890 | 1 h, 250 GB | | brain01, 2026-11-03 |
| 4 | 16618 | 16618 | conforms | 81.7 | 2623 | 1 h, 250 GB | | brain01, 2026-11-03 |

### context-free twins online, Java port (JMC) against the Rust implementation's export: pair **counting-bug**, model bag, expected violation

| size | $G_{spec}$ | $G_{impl}$ | $G_x$ | verdict | T | M | tier | first miss ms | run |
|---|---|---|---|---|---|---|---|---|---|
| 2 | 76 | 60 | 12 | violation | 3.5 | 158 | 1 h, 250 GB | 2069 | brain01, 2026-11-02 |
| 3 | 1690 | 826 | 324 | violation | 11.7 | 1859 | 1 h, 250 GB | 5063 | brain01, 2026-11-03 |
| 4 | 72346 | 16618 | 9072 | violation | 83.1 | 2576 | 1 h, 250 GB | 5445 | brain01, 2026-11-03 |

**Not run (context-free twins online, Java port (JMC) against the Rust implementation's export).** planned but never scheduled (grid points beyond the ladder): 5 bag correct, 6 bag counting-bug, 6 bag correct, 7 bag counting-bug.

## context-free twins online, Rust implementation (TraceForge) against the Java port's export

### context-free twins online, Rust implementation (TraceForge) against the Java port's export: pair **correct**, model bag, expected conforms

| size | $G_{spec}$ | $G_{impl}$ | verdict | T | M | tier | first miss ms | run |
|---|---|---|---|---|---|---|---|---|
| 2 | 76 | 76 | conforms | 3.6 | 158 | 1 h, 250 GB | | brain01, 2026-11-02 |
| 3 | 1690 | 1690 | conforms | 8.7 | 636 | 1 h, 250 GB | | brain01, 2026-11-02 |
| 4 | 72346 | 72346 | conforms | 288.3 | 3427 | 1 h, 250 GB | | brain01, 2026-11-02 |

### context-free twins online, Rust implementation (TraceForge) against the Java port's export: pair **counting-bug**, model bag, expected conforms

| size | $G_{spec}$ | $G_{impl}$ | verdict | T | M | tier | first miss ms | run |
|---|---|---|---|---|---|---|---|---|
| 2 | 60 | 60 | conforms | 3.1 | 158 | 1 h, 250 GB | | brain01, 2026-11-02 |
| 3 | 826 | 826 | conforms | 5.9 | 392 | 1 h, 250 GB | | brain01, 2026-11-02 |
| 4 | 16618 | 16618 | conforms | 64.1 | 2325 | 1 h, 250 GB | | brain01, 2026-11-02 |

**context-free twins online, Rust implementation (TraceForge) against the Java port's export: pair counting-bug, model bag, expected violation**

| size | $G_{spec}$ | $G_{impl}$ | $G_x$ | verdict | T | M | tier | first miss ms | run |
|---|---|---|---|---|---|---|---|---|---|
| 2 | 76 | 60 | 12 | violation | 3.4 | 158 | 1 h, 250 GB | 2 | brain01, 2026-11-02 |
| 3 | 1690 | 826 | 324 | violation | 8.6 | 636 | 1 h, 250 GB | 3 | brain01, 2026-11-02 |
| 4 | 72346 | 16618 | 9072 | violation | 287.6 | 3427 | 1 h, 250 GB | 7 | brain01, 2026-11-02 |

**Not run (context-free twins online, Rust implementation (TraceForge) against the Java port's export).** planned but never scheduled (grid points beyond the ladder): 5 bag correct, 6 bag counting-bug, 6 bag correct, 7 bag counting-bug.